

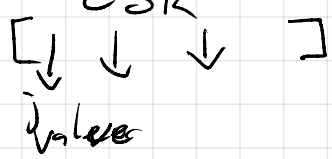
COO

i

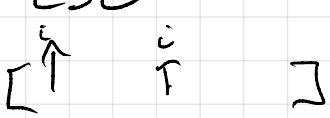
j

values

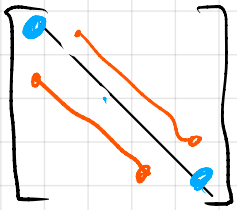
CSR



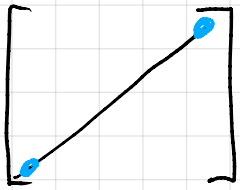
CSC



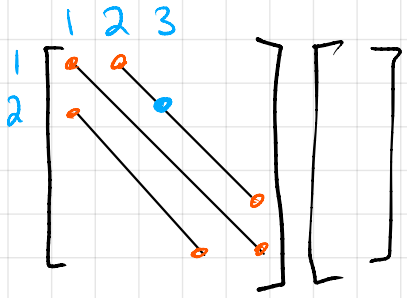
Structure exploitation



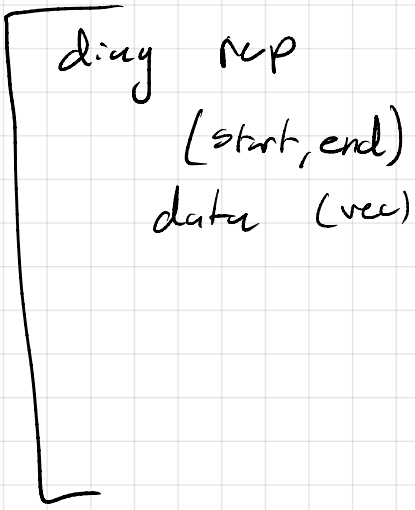
$i = j$



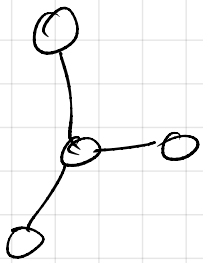
$i = N - j$

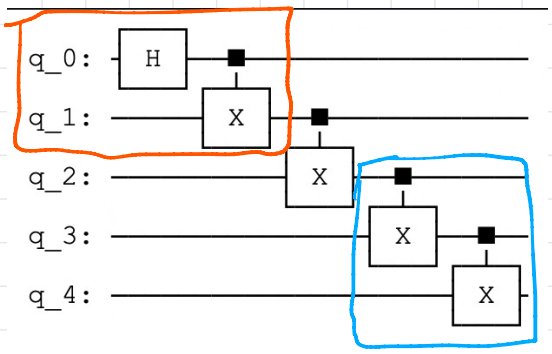


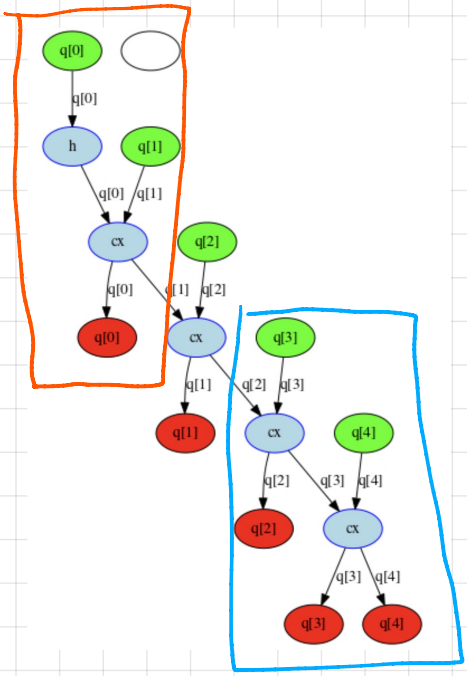
[2, 3]



$H \otimes H$







$$(CNOT)(H \otimes I)(q_0 \otimes q_1)$$

GOAL: Memory Efficiency
(not contraction choices)

$$[(CNOT)[(H \otimes I)(H \otimes I)(H \otimes I)(H \otimes I)](q_0 \otimes q_1)]$$

New struct:

start i, j
end k, l
data : [, , , , , , ,]

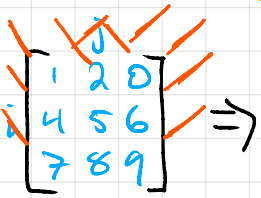
complex "integers"

a + bi w/ a, b ∈ ℤ

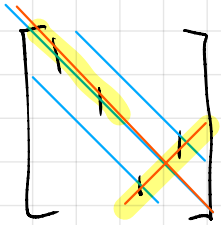
a, b ∈ {-1, 0, 1}

Most existing pkgs use numpy

Dense ⇒ ~~struct?~~ ↳ Class?



Assume principal diagonals



$H H^*$

do we need constants?

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$\sqrt{1+1} = \sqrt{2}$

